

DMIMS Developer Implementation Checklist

Supporting Document 2

1. Purpose

This checklist helps the developer build DMIMS step by step without missing key features.

The master blueprint explains what the system must do.

This checklist explains the order of implementation and what must be verified.

Phase 1 — Project Setup

1.1 Environment

Confirm:

Laravel installed
Filament installed
MySQL/MariaDB connected
.env configured
APP_KEY generated
Composer dependencies installed
Node dependencies installed
Vite assets build correctly

Commands:

composer install
npm install
php artisan key:generate
php artisan migrate
npm run build
php artisan serve

Acceptance:

Application loads
Filament login loads
No migration error
No asset build error

Phase 2 — Authentication and Roles

2.1 Install Permission System

Use:

spatie/laravel-permission

Create roles:

Datamation Super Admin
Datamation Management
Company Admin
Company Supervisor
Stock Inventory User
Document Tracking User
Viewer

Acceptance:

Roles seeded
Permissions seeded
Super Admin can access all
Management user is read-only
Company users cannot access platform admin pages

Phase 3 — Customer Company Management

Build:

customers table
Customer model
CustomerResource
Company status handling

Features:

Create customer
Edit customer
Suspend customer
Reactivate customer
Archive customer
View users
View subscription
View license
View billing
View audit logs

Acceptance:

Super Admin can create Company A and Company B
Company code must be unique
Suspended company cannot perform operations
Archived company hidden from normal dropdowns

Phase 4 — User Management

Build:

customer_id in users
is_platform_user in users
UserResource
Role assignment
User status

Acceptance:

Platform user has customer_id NULL
Company user has customer_id
Company Admin sees only own users
Company users cannot create platform users
Inactive users cannot login

Phase 5 — Customer Isolation

Apply customer filtering to:

products
categories
locations
boxes
document_files
stock_movements
document_movement_logs
barcode_registry
barcode_scan_logs
billing_records
subscriptions
licenses
audit_logs
notifications

Acceptance:

Company A user cannot see Company B data
Company B user cannot see Company A data
Direct URL access to another company record is blocked
Super Admin can see all

Phase 6 — Module Control

Build:

modules
customer_modules
ModuleResource
Customer module assignment
EnsureModuleEnabled middleware

Acceptance:

Disabled module hidden from menu

Disabled module blocked by URL
Disabled module blocked in service/action
Customer module settings work

Phase 7 — Subscription Control

Build:

subscription_plans
customer_subscriptions
subscription_logs
SubscriptionService

Subscription controls:

Plan
Modules
Limits
Valid dates
Billing cycle

Required checks:

max_users
max_products
max_document_files
max_boxes
valid_from
valid_to
grace_period_days
enabled_modules

Acceptance:

Subscription plan can be created
Customer subscription can be assigned
Renewal works
Grace period works
Expired subscription blocks operations unless license allows view/export
Subscription logs created
Subscription limits enforced

Phase 8 — License Control

Build:

licenses
license_logs
LicenseService
EnsureLicenseAllowsAccess middleware

License controls:

active
suspended
revoked
expired
technical access mode

Acceptance:

Active license allows normal access
Suspended license allows view/export only
Expired license allows view/export only
Revoked license blocks access
License logs created
License status shown on dashboard

Phase 9 — AccessControlService

Build:

AccessControlService

Must combine:

User status
Company status
Subscription status
License status
Module access

Permission
Usage limits

Required methods:

canLogin(\$user)
canView(\$user, \$module)
canExport(\$user, \$module)
canPerformOperationalAction(\$user, \$module, \$action)
getEffectiveLimits(\$customerId)
getEffectiveAccessMode(\$customerId)

Acceptance:

All major actions use AccessControlService
No important action depends only on hidden UI buttons
Direct URL bypass fails

Phase 10 — Manual Billing and Payment

Build:

billing_records
billing_payments
billing_logs
BillingService
PaymentService
BillingRecordResource
BillingPaymentResource

Manual payment only:

No online payment gateway
Super Admin updates status manually
Management role can view summaries only
Customer users cannot edit billing

Acceptance:

Billing record can be created
Invoice number can be stored
Manual payment can be added

Paid amount updates
Balance amount updates
Partial payment works
Overdue status works
Billing logs created
Payment update audited

Phase 11 — Shared Location Management

Build:

location_types
locations
LocationResource
LocationService

Rules:

Locations are shared by stock and document modules
Do not create fake external locations
Locations appear under both menus

Acceptance:

Location tree works
Nested locations work
Location barcode works
Stock can use locations
Boxes can use locations
External destinations are not stored as locations

Phase 12 — Stock Inventory

Build:

categories
products

product_location_stocks
stock_movements
stock_alerts
StockMovementService

Pages:

Categories
Products
Stock Receive-In
Stock Internal Transfer
Stock Out
Stock Adjustment
Stock Movements
Stock Reports

Acceptance:

Product can be created
SKU unique per company
Barcode unique per company
Receive-in increases stock
Stock out decreases stock
Stock out cannot exceed available quantity
Internal transfer moves stock between locations
Adjustment requires reason
Movement logs created
Low stock alert works

Phase 13 — Document Tracking

Build:

document_types
boxes
document_files
document_movement_logs
DocumentMovementService

Pages:

Boxes

Document Files
File Receive-In
File Internal Transfer
File Move-Out
File Return
Box Receive-In
Box Internal Transfer
Box Move-Out
Box Return
Movement Logs
Document Reports

Acceptance:

Box receive-in works
File receive-in works
File assigned to box
File transfer works
Box transfer works
File move-out captures external destination
Box move-out captures external destination
File return works
Box return works
Moved-out file cannot transfer internally
Closed box cannot receive files
Movement logs immutable

Phase 14 — Movement Architecture

Implement three movement models:

Internal Transfer
External Receive-In
External Move-Out

Acceptance:

Internal transfer requires DMIMS source and destination
External receive-in uses source_type/source_name
External move-out uses destination_type/destination_name
No fake locations are required
File last_internal_box_id preserved
Box last_internal_location_id preserved

Overdue returns detectable

Phase 15 — Barcode System

Build:

barcode_registry
barcode_scan_logs
BarcodeService
ScannerService
ScannerPage
Barcode printing

Acceptance:

Product barcode works
Location barcode works
Box barcode works
File barcode works
Unknown barcode handled
Barcode from another customer blocked
Scan logs created
Print count updated

Phase 16 — Reports and Analytics

Build platform reports:

Customer Report
Subscription Report
License Report
Billing Report
Payment Report
Outstanding Balance Report
Audit Log Report
Module Usage Report

Build stock reports:

Inventory Summary

Low Stock
Stock Movement
Stock Receive-In
Stock Out
Stock Internal Transfer
Stock Value

Build document reports:

File Master
Box Master
Files by Box
Boxes by Location
File Movement History
Box Movement History
File Move-Out
Box Move-Out
Overdue Return
External Movement

Acceptance:

Company users see own data only
Super Admin can filter by customer
Management role can view analytics only
Exports work
Export actions audited

Phase 17 — Audit Logs

Build:

AuditService
audit_logs

Audit:

Login/logout
Failed login
User changes
Company changes
Subscription changes

License changes
Billing/payment changes
Stock movements
Document movements
Barcode scans/prints
Imports/exports
Backup/restore

Acceptance:

Audit log records old/new values
Company users cannot edit audit logs
Super Admin can view all audit logs
Management can view summaries only if allowed

Phase 18 — Notifications

Build:

notifications
NotificationService

Notify for:

Low stock
Subscription near expiry
License near expiry
Billing overdue
Payment updated
File return overdue
Box return overdue
Import failed
Export completed

Acceptance:

Notifications appear to correct users
Company notifications stay within same customer_id
Platform notifications go to Super Admin

Phase 19 — Import / Export

Build import for:

- Products
- Opening stock
- Locations
- Boxes
- Document files

Rules:

- Preview before import
- Validate rows
- Show invalid rows
- Download error file
- Respect limits
- Audit import
- Block import in view-only mode

Acceptance:

- Valid import succeeds
- Invalid import shows errors
- Duplicate SKU blocked
- Duplicate barcode blocked
- Import respects subscription/license limits

Phase 20 — PWA

Build:

- manifest.json
- service-worker.js
- offline page
- PWA icons
- mobile scanner layout
- install prompt

Acceptance:

- PWA install works

Offline fallback works
Mobile scanner page usable
Tables responsive
HTTPS enforced

Phase 21 — Backup / Restore

Build:

BackupService
Backup command
Restore SOP
Backup logs

Backup must include:

Database
Uploaded files
Import files
Export files if needed
Barcode templates

Acceptance:

Manual backup works
Scheduled backup works
Backup copied to NAS/local target
Restore test succeeds
Backup action audited

Phase 22 — Production Deployment

Prepare:

Ubuntu Server
Nginx
PHP-FPM
MySQL/MariaDB
Composer
Node.js

Laravel queue worker
Supervisor
Cron scheduler
Cloudflare Tunnel
SSL
Production .env
Backup script

Production **.env**:

```
APP_NAME="DMIMS"  
APP_ENV=production  
APP_DEBUG=false  
APP_URL=https://dmims.datamationgroup.com
```

```
SESSION_SECURE_COOKIE=true  
SESSION_DRIVER=database  
CACHE_STORE=file  
QUEUE_CONNECTION=database  
TRUSTED_PROXIES=*
```

Acceptance:

```
APP_DEBUG=false  
HTTPS works  
Login works via public URL  
Session does not break behind Cloudflare Tunnel  
Queue worker running  
Scheduler running  
Backup running
```

Final Go-Live Checklist

Before production:

All migrations run successfully
All seeders run successfully
Super Admin login works
Management login read-only works
Customer company created
Company user created
Company isolation verified
Subscription control verified

License control verified
Manual billing/payment verified
Stock workflows verified
Document workflows verified
Barcode workflows verified
Reports verified
Audit logs verified
PWA verified
Backup/restore verified
Security checks passed

Do not go live until all items are tested.